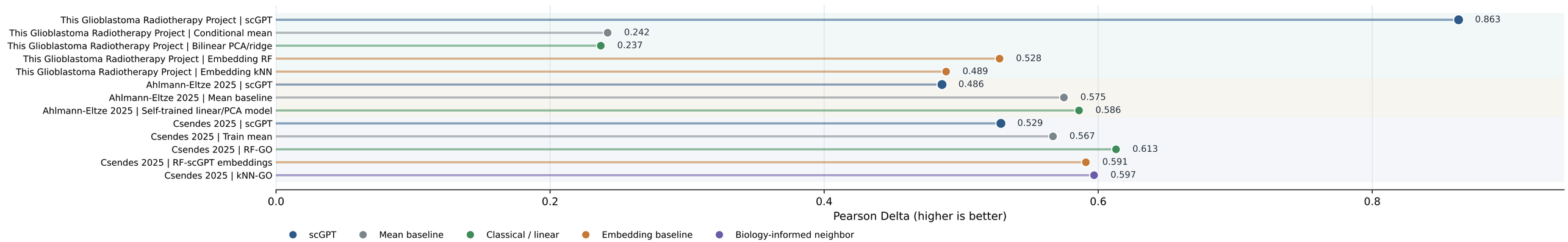
Literature Benchmark Context for This Glioblastoma Radiotherapy Project

A. Benchmark Values Used for Report Positioning

Study	Dataset / setting	Model	Pearson	Pearson DE	Pearson Delta	Pearson Delta DE	Interpretation
This Glioblastoma Radiotherapy Project	Glioblastoma, ctrl vs IR-24h / IR-1week	scGPT	0.974	0.862	0.863	0.904	Values suggest the most favorable performance for scGPT among the listed project models.
This Glioblastoma Radiotherapy Project	Same	Conditional mean	0.552	0.423	0.242	0.513	Condition-average response appears limited for this setting.
This Glioblastoma Radiotherapy Project	Same	Bilinear PCA/ridge	0.551	0.411	0.237	0.503	Low-rank linear structure appears limited in this benchmark.
This Glioblastoma Radiotherapy Project	Same	Embedding RF	0.684	0.694	0.528	0.735	Values suggest improvement over mean-style baselines, while remaining below scGPT in this summary.
This Glioblastoma Radiotherapy Project	Same	Embedding kNN	0.655	0.699	0.489	0.736	Values suggest local similarity is informative, while still below scGPT here.
Ahlmann-Eltze et al., 2025	Adamson, Replogle K562/RPE1; unseen single-gene perturbations	scGPT	0.975	Not reported	0.486	Not reported	scGPT appears lower than simple baselines on the reported single-gene task.
Ahlmann-Eltze et al., 2025	Same	Mean baseline	0.978	Not reported	0.575	Not reported	Mean baseline shows higher Pearson Delta than scGPT in this summary.
Ahlmann-Eltze et al., 2025	Same	Self-trained linear/PCA model	0.979	Not reported	0.586	Not reported	Values point toward the self-trained linear/PCA model as most favorable among these rows.
Csendes et al., 2025	Adamson, Norman, Replogle K562/RPE1; CRISPR Perturb-seq	scGPT	0.985	0.950	0.529	0.666	The reported values do not suggest consistent scGPT advantage over baselines on delta metrics.
Csendes et al., 2025	Same	Train mean	0.986	0.933	0.567	0.618	Mean baseline shows higher Pearson Delta than scGPT in this summary.
Csendes et al., 2025	Same	RF-GO	0.987	0.942	0.613	0.669	Values point toward GO-informed RF as the most favorable listed Csendes baseline.
Csendes et al., 2025	Same	RF-scGPT embeddings	0.987	0.940	0.591	0.637	scGPT embeddings may be useful for RF, though not clearly best overall.
Csendes et al., 2025	Same	kNN-GO	0.987	0.944	0.597	0.659	kNN with biological features appears competitive.

B. Pearson Delta Illustration Across Studies



Sources: Ahlmann-Eltze et al. 2025 Source Data Fig. 2; Csendes et al. 2025 Supplementary Table 2; project summary values for this GB radiotherapy benchmark. Cross-study comparisons are descriptive because datasets and perturbation settings differ.